Wave Optics

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Huygen's Wave Theory

Nature of light

Ray optics assumes that light travels in straight line. But from this chapter, we get to know that light does not always travel in straight line, indeed it behaves as a wave that interfere, diffract ever under goes polarization. The branch of physics that deals with wave nature of light is called **'wave optics'**.

Initially, Newton developed the **corpuscular theory model** that was originally given by Descartes in 1637. As per this model, a luminous body emits a stream of particles in all directions. The particles are assumed to be very tiny.

This theory could explain the phenomena of reflection and refraction, but it could not explain phenomena like interference, polarization and diffraction.

After Newton, **Christian Huygen in 1678 proposed wave nature of light** and satisfactorily demonstrated law of reflection, refraction, the rectilinear propagation of light and the fact that light could propagate through vacuum.

Huygen's theory got acceptance when Thomas Young performed his famous interference experiments in 1801. Following his experiment, many experiments were carried out involving the interference and diffraction of light waves. Around the middle of Nineteenth Century, the wave theory gained well acceptance.

At that time, it was a firm belief that no wave could travel without a medium, so Huygen's wave theory failed to explain the propagation of light in vacuum, but later when Maxwell propounded his famous electromagnetic theory, then the nature of light waves was understood properly.

Maxwells electromagnetic wave theory supported the wave nature of light.

Max Planks quantum theory supported particle nature of light

Einstein's photo electric effect supported particle nature of light.

Finally, it was concluded that light has **dual nature- particle as well as wave**.

In this chapter, we will study about wave nature of light for which we will follow Huygens theory

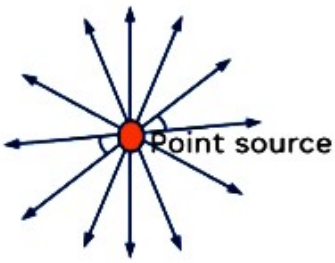
Huygens's Wave Theory

⇒ Body emits light in the form of waves in all directions.

⇒ Each point source of light is a centre of disturbance and speed of disturbance in vacuum is $c = 3 \times 10^8$ m/s.

⇒ Velocity of disturbance in a medium is, $v = \frac{c}{\mu}$

Here μ is refractive index of the medium



⇒ The locus of all the medium particles which are vibrating in the same phase at any given time instant is called the wave front.

⇒ Each point on a wave front is a source of new disturbance and called secondary wavelets.

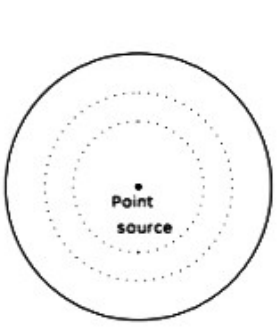
⇒ Each ray takes same time to reach from one wavefront to another wavefront.

⇒ Every single point on a wavefront acts as a source of new secondary wavelets. The secondary wavelets are spherical in shape and they travel with the speed of light in all directions in the medium.

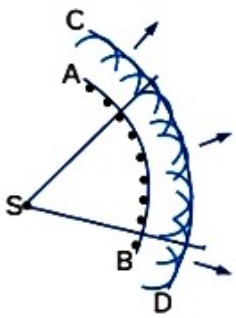
⇒ The envelope enclosing all the tangents drawn at the secondary wavelets in forward direction gives the position of the new wavefronts.

⇒ Huygens assumed that energy does not flow backwards when wave travels in forward direction.

⇒ To plot a wavefront, draw arc of length vt from secondary wavelet and tangent to these arcs in forward direction gives new wavefront.

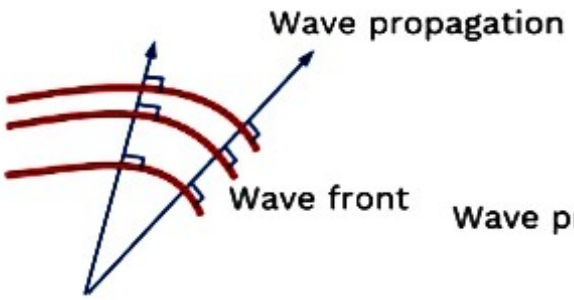


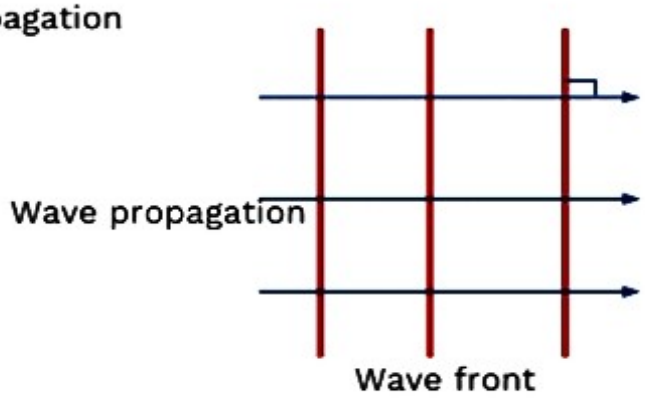
Wave front

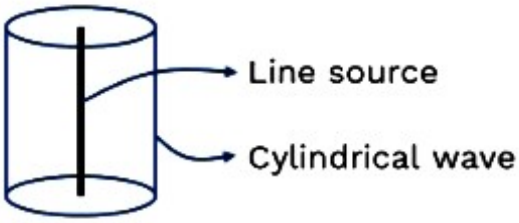


Huygens' Principle:
Each wavefront is the envelop of the wavelets. Each point on a wavefront acts as an independent source to generate wavelets for the next wavefront. AB and CD are two wavefronts.

⇒ In homogeneous medium, direction of wave propagation is always perpendicular to wave fronts.

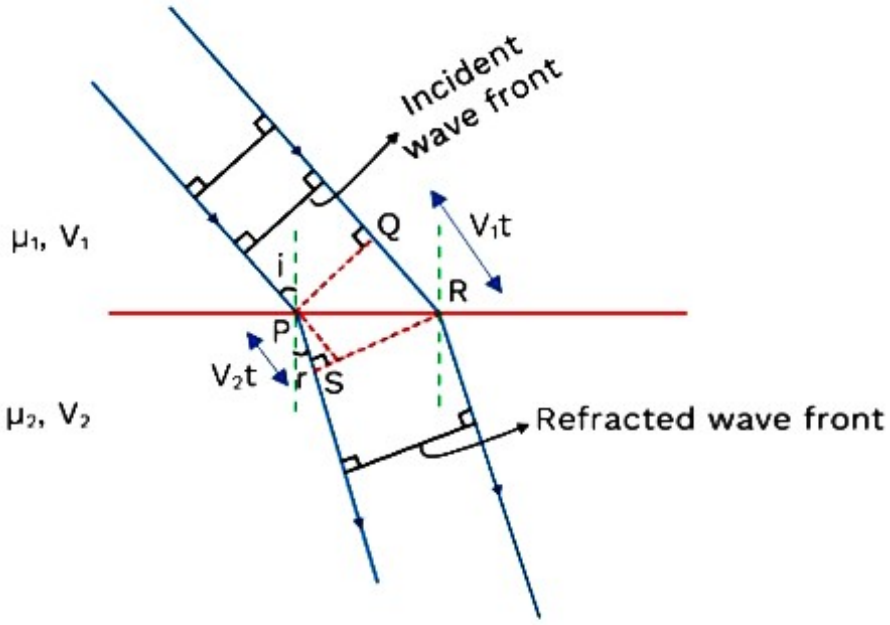






Refraction and Reflection by Huygen's theory

a. Refraction of a plane wave



Distance moved by light in time t in medium 1, QR = V_1t
And, in medium 2, PS = V_2t

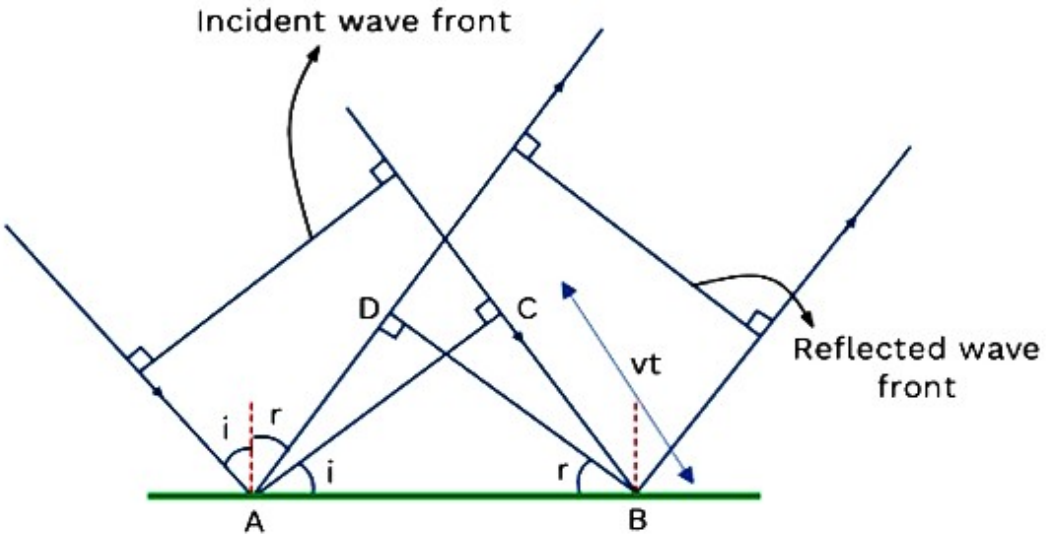
Let's assume PR is x.

In ΔACB, $\sin i = \frac{V_1 t}{x}$, In ΔADB, $\sin r = \frac{V_2 t}{x}$

$$\frac{\sin i}{\sin r} = \frac{V_1}{V_2} = \frac{C}{\mu_1}\frac{\mu_2}{C} = \frac{\mu_2}{\mu_1}$$

$\boxed{\mu_1 \sin i = \mu_2 \sin r}$, which is Snell's law.

b. Reflection of a plane wave



Distance travelled by incident light in time t, BC = vt

Distance travelled by reflected light in same time, AD = vt

In triangle ACB, $\sin i = \frac{BC}{x} = \frac{V t}{x}$

In triangle BDA $\sin r = \frac{AD}{x} = \frac{V t}{x}$

Q. **What is the angle made by the refracted wavefront with horizontal when a plane wavefront is incident on a boundary separating air and another medium of refractive index $\mu = \frac{3}{2}$ while making an angle of 37° with the horizontal horizontal.**

Sol. As the incident wavefront makes 37° with horizontal, then the incident ray will make 37° with the normal to the plane as wavefront and ray are perpendicular to each other.

By Snell's law,

$$\frac{\sin 37^\circ}{\sin r} = \frac{3}{2}$$

$$\sin r = \frac{2}{3} \times \frac{3}{5} = \frac{2}{5}$$

$$r = \sin^{-1}\left(\frac{2}{5}\right)$$

which is same as angle of refractive wavefront with horizontal.

Q. **Yellow light with wavelength 0.5 μm in air suffers refraction in a medium in which velocity of light is 2×10^8 m/s. Then the wavelength of the light in the medium would be.**

Sol. $\frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2}$

Here, $\lambda_1 = 0.5\mu m$, $v_1 = 3 \times 10^8 \, m/s$

Let, $\lambda_2 = x$, $v_2 = 2 \times 10^8 \, m/s$

$$\frac{0.5}{x} = \frac{3 \times 10^8}{2 \times 10^8}$$

$x = 0.33$ μm

$\sin i = \sin r$

$\angle i = \angle r$, which is law of reflection.

Note: The frequency remains unchanged.

Doppler's effect on light

When there is a relative motion between light source and observer then there is a frequency shift in observed light.

Case 1 - Red Shift :- This shift in the frequency of light observed when light source moves away from observer the wavelength of light decreases and shifted towards wavelength of red light.

Case 2 – Blue Shift :- This shift in the frequency of light observed when light source moves toward the observer. The frequency of light shifted to frequency of blue light.
The apparent shift in frequency and wavelength are found as described below:
Case-1 : Source and observer approach each other

$$f_A = f\sqrt{\frac{c+v}{c-v}}$$



Here, c is the speed of light
v is the relative speed of approach
f is the frequency of wave emitted by source
f_A is the observed frequency

Now, if v << c, then by binomial expansion, we have

$$f_A = f\left[1+\frac{v}{c}\right]^{1/2}\left[1-\frac{v}{c}\right]^{-1/2}$$

$$\Rightarrow \quad f_A = f\left[1+\frac{v}{2c}\right]\left[1+\frac{v}{2c}\right] \quad \left(\text{neglecting higher powers of } \frac{v}{c}\right)$$

$$\Rightarrow \quad \boxed{f_A = f\left[1+\frac{v}{c}\right]}$$

Case-2: If the source or the observer recede from each other.

$$f_R = f\sqrt{\frac{c-v}{c+v}}$$



which again reduces to $f_R = f\left[1-\frac{v}{c}\right]$ for v << c, here f_R represents frequency of recession.

The same relationship can be modified to get apparent wavelength.

$$\lambda = \frac{c}{f}$$

$$\Rightarrow \frac{c}{\lambda_A} = \frac{c}{\lambda}\left[1+\frac{v}{c}\right]$$

$$\Rightarrow \lambda_A = \lambda\left[1+\frac{v}{c}\right]^{-1}$$

$$\Rightarrow \lambda_A = \lambda\left[1-\frac{v}{c}\right]$$

$$\Rightarrow \lambda_A - \lambda = -\lambda\frac{v}{c}$$

$$\boxed{-\frac{\Delta\lambda}{\lambda} = \frac{v}{c}}$$

Note that in the above formulae, v is the component of the source velocity along the line joining the observer to the source relative to the observer.
v is positive when source moves relatively away from observer and v is negative when source moves towards the observer.
The Doppler effect for the light is useful in astronomy. It is used to measure the radial velocities of distant galaxies.

Q. **What should be the velocity of a star wrt earth so that the light beam of wavelength 564 nm is observed to be 564.8 nm?**

Sol. Now we have,

$$\frac{\Delta\nu}{\nu} = -\frac{\Delta\lambda}{\lambda} = \frac{-V_{radial}}{c}$$

Now, $\Delta\lambda = (564.8 - 564.0)nm = 0.8nm$

Now, $v_{radial} = \frac{c\Delta\lambda}{\lambda} = \frac{3\times10^8 \times 0.8nm}{564nm}$

$= 4.255 \times 10^5$ m/s

$\cong 425.5km/s$

Since, v comes out to be positive, hence the star is moving away from earth.

Q. **A galaxy moving with speed 300 km/s shows blue shift. At what wavelength sodium line at 589.0 nm will be observed?**

Sol. Here, $v_{radial} = -300$ km/s $= -3 \times 10^5$ m/s
Negative sign comes because we observe blue shift i.e. galaxy is approaching us.

$$\text{Now, } -\frac{\Delta\lambda}{\lambda} = -\frac{v_{radial}}{c}$$

$$\Rightarrow -\frac{\Delta\lambda}{589.0nm} = \frac{-\left(-3\times10^5\right)}{\left(3\times10^8\right)}$$

$$\Rightarrow \Delta\lambda = -(589.0)\times10^{-3}nm$$

$= -0.589$ nm

Hence, the observed wavelength will be (589.0 – 0.589) nm = 588.411 nm.

Interference of Light

Coherent and Incoherent sources of wave:

The sources of light which emit waves with zero or a constant phase difference and having same frequency are called coherent sources.

Two different light sources can never produce coherent waves.

Incoherent source: Those sources which have different frequencies, or their phase difference changes with time.

Note: The sources which are lying on the same wavefront are coherent in nature because phase difference between them is zero

$$y_1 = A_1 \sin(\omega t + kx)$$

$$\left.\begin{aligned} y_1 &= A_1 \sin(\omega t + kx) \\ y_2 &= A_2 \sin(\omega t + kx) \end{aligned}\right\} \text{ (Coherent sources) Phase difference } = 0 \text{(constant)}$$

$$\left.\begin{aligned} y_1 &= A_1 \sin(\omega t + kx + 30°) \\ y_2 &= A_2 \sin(\omega t + kx + 60°) \end{aligned}\right\} \text{ (Coherent sources) Phase difference } = 30° \text{ (constant)}$$

Principle of superposition

For two superimposing waves at any point in space, the resultant wave displacement is equal to the sum of the displacements of the individual waves.

Interference:

In interference, two or more waves are superimposed to form a new resultant wave.
Consider a homogeneous medium in which two sinusoidal waves are interfering.

$$y_1 = A_1 \sin \omega t$$

$$y_2 = A_2 \sin(\omega t + \phi)$$

Then by principle of superposition, resultant wave equation is given by:

$$y = y_1 + y_2$$

$$= A_1 \sin \omega t + A_2 \sin(\omega t + \phi)$$

$$= A_1 \sin \omega t + A_2[\sin \omega t \cos \phi + \cos \omega t \sin \phi]$$

$$= (A_1 + A_2 \cos \phi) \sin \omega t + A_2 \sin \phi \cos \omega t$$

Let amplitude of resultant wave is A and phase constant is δ.

$$A_1 + A_2 \cos \phi = A \cos \delta \quad \text{..... (i)}$$

$$A_2 \sin \phi = A \sin \delta \quad \text{..... (ii)}$$

$(i)^2 + (ii)^2$

$$(A_1 + A_2\cos\phi)^2 + (A_2\sin\phi)^2 = A^2\left(\cos^2\delta + \sin^2\delta\right)$$

$$A^2 = A_1^2 + A_2{}^2 + 2A_1A_2\cos\phi$$

Now we know that , $I \propto A^2$

So, $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

Case(i) Intensity will be maximum.

When $\cos\phi = 1$

$\phi = 2n\pi \quad (n \in I)$ or path difference

$$\Delta x = \frac{\lambda}{2\pi}\phi = n\lambda$$

$$\boxed{I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2}$$

The interference when the resultant intensity is maximum is known as constructive interference.

Case(ii) Intensity will be minimum.

When $\cos\phi = -1$

$\phi = (2n + 1)\pi$ or Path difference $\Delta x = \frac{\lambda}{2\pi}\phi = (2n + 1)\frac{\lambda}{2\pi}$

$$\boxed{I_{min} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2}$$

The interference when the resultant intensity is minimum is called **destructive interference**.

Note: When interfering sources are incoherent, then Intensity of resultant wave is given by sum of individual intensities of waves and no interference pattern is observed.

Q. Calculate the ratio of maximum to minimum intensity when the ratio of amplitude of two waves is 4:3?

Sol. Let amplitudes 4x and 3x respectively. At maxima, amplitudes add up giving 7x. At minima, amplitudes subtract giving x.

Intensities ratio = $\frac{49}{1}$, as the Intensity is square of amplitude.

Q. **Two waves having intensity in the ratio 25: 4 produce interference. Find the ratio of maximum to minimum intensity.**

Sol.

$$\frac{I_{max}}{I_{min}} = \left(\frac{\sqrt{\frac{I_1}{I_2}}+1}{\sqrt{\frac{I_1}{I_2}}-1}\right)^2$$

$$= \left(\frac{\sqrt{\frac{25}{4}}+1}{\sqrt{\frac{25}{4}}-1}\right)^2 = \frac{49}{9}$$

Q. **Find the amplitude of the resultant wave when two wave represented by $y_1 = 4\sin\omega t$ and $y_2 = 3\sin\left(\omega t + \frac{\pi}{3}\right)$ interfere at a point.**

Sol. $\phi = \pi/3, \quad a_1 = 4, \quad a_2 = 3$

So,

$$A = \sqrt{a_1^2 + a_2^2 + 2a_1 \cdot a_2 \cos\phi} \Rightarrow A \approx 6$$

Q. **Two coherent sources each emitting light of intensity I_0 interfere in a medium at a point where phase difference between them is $\frac{2\pi}{3}$. Find the resultant intensity at the given point.**

Sol. Given that, $\phi = \frac{2\pi}{3}$

and, $I_{res} = 4I_0\cos^2\left(\frac{\phi}{2}\right)$

$$= 4I_0\cos^2\left(\frac{\pi}{3}\right) = I_0$$

Q. **Find the maximum and minimum possible intensity when two source having intensity I_0 and $4I_0$ interfere at a point in a medium.**

Sol. $I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$

Here, $I_1 = I_0$ and, $I_2 = 4I_0$

$\therefore \quad I_{max} = \left(\sqrt{I_0} + \sqrt{4I_0}\right)^2 = 9I_0$

and, $I_{min} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2 = I_0$

Q. **Find the resultant intensity of light produced by interference of light from to incoherent sources having intensity I_0 and $3I_0$**

Sol. Any sustained interference cannot be observed as the sources are incoherent. We observe an average intensity,

$\langle I \rangle = \left\langle I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi \right\rangle$

Now, $\langle \cos\phi \rangle = 0$ in one complete cycle.

$\therefore \quad \langle I \rangle = I_1 + I_2$, which is $4I_0$.

Some important formulae

1. During constructive interference, resultant amplitude is becomes maximum and is given by $R_{max} = A + B$.
2. During destructive interference, resultant amplitude is becomes minimum, and is given by $R_{min} = |A - B|$

$$\frac{R_{max}}{R_{min}} = \frac{A+B}{|A-B|}$$

3. As $I_{max} \propto R_{max}^2$, $I_{min} \propto R_{min}^2$

$$\Rightarrow \frac{I_{max}}{I_{min}} = \left(\frac{A+B}{A-B}\right)^2 = \left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}}\right)^2$$

4. If w_1 and w_2 are widths of two slits from which intensities of light are I_1 and I_2, then,

$$\frac{w_1}{w_2} = \frac{I_1}{I_2} = \frac{a^2}{b^2}$$

Q. **Two coherent sources having a ratio of intensities as 81:1 produce interference. Find the ratio of maxima and minima in the interference pattern observed.**

Sol. Here, $\frac{I_1}{I_2} = \frac{a^2}{b^2} = \frac{81}{1}$

$\therefore \quad \frac{a}{b} = \frac{9}{1}$

$\Rightarrow \quad a = 9b$

In the interference pattern

$$\frac{I_{max}}{I_{min}} = \frac{(a+b)^2}{(a-b)^2} = \frac{(9b+b)^2}{(9b-b)^2} = \frac{100b^2}{64b^2} = \frac{25}{16}$$

Q. **Light waves from coherent sources having intensities I and 4I cross each other at a point with a phase difference of 60°. What is the resultant intensity at the point?**

Sol. Here, $I_1 = I$ and $I_2 = 4I$, $\phi = 60°$

$$I_R = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

$$= I + 4I + 2\sqrt{I \times (4I)}\cos\phi$$

$$= 5I + 2I\cos 60°$$

$$= 5I + \frac{2I}{2} = 6I$$

Q. **When n identical waves each having an intensity I_0 interfere, then find the maximum intensity observed when,**
(a) All the sources are coherent
(b) All the sources are incoherent

Sol.

Intensity of each of n waves = I_0

For coherent interference,

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

$$I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos 0° = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$$

We have n identical waves instead of 2 waves, so for n waves,

$$I_{max} = \left(\sqrt{I_0} + \sqrt{I_0} + \ldots\ldots \text{n times}\right)^2 = \left(n\sqrt{I_0}\right)^2 = n^2 I_0$$

For interference of incoherent sources, phase difference ϕ varies randomly with time.

$(\cos\phi)_{av} = 0, \quad I_{max} = I_1 + I_2$

So, for n incoherent sources,

$I_{max} = I_0 + I_0 + \ldots\ldots \text{n times}$

$= nI_0.$

Conservation of energy and Interference

In the interference pattern, if we take $I_{max} = k(a+b)^2$ and, $I_{min} = k(a-b)^2$, then average intensity of light in the interference pattern

$$I_{avg} = \frac{I_{max} + I_{min}}{2} = \frac{k(a+b)^2 + k(a-b)^2}{2}$$

$$I_{avg} = \frac{2k\left(a^2 + b^2\right)}{2} = k\left(a^2 + b^2\right)$$

If there were no interference, intensity of light from two sources at every point on the screen would be $I = I_1 + I_2 = ka^2 + kb^2 = k\left(a^2 + b^2\right)$, which is the same as I_{avg} in the interference pattern.

It shows that energy is neither created nor destroyed in interference process. It is simply redistributed.

Q. **For two sources having slightly different intensities given by I and (I + δI), such that δI << I, find the intensity at maxima and at minima.**

Sol. Using the relation, $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

$$I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos 0°$$

$$= I + (I + \delta I) + 2\sqrt{I(I + \delta I)}$$

As, $\delta I << I$, therefore,

$I_{max} = I + I + 2I = 4I$

Again from the relation,, $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

$I_{min} = I + (I + \delta I) + 2\sqrt{I(I + \delta I)}\cos 180°$

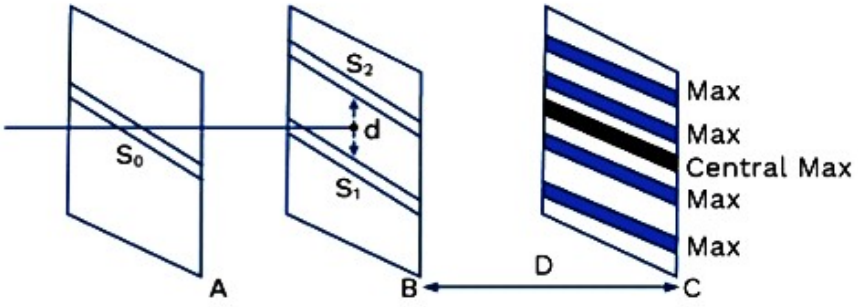
$$= 2I + \delta I - 2I\left(1 + \frac{\delta I}{I}\right)^{\frac{1}{2}} = 2I + \delta I - 2I\left[1 + \frac{1}{2}\frac{\delta I}{I} + \frac{\frac{1}{2}\left(\frac{1}{2} - 1\right)}{2!}\left(\frac{\delta I}{I}\right)^2\right]$$

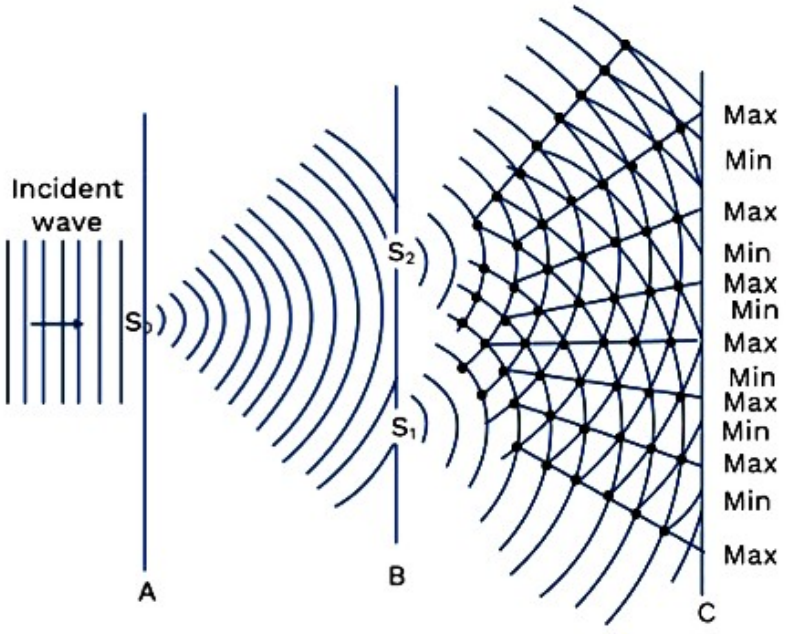
$$= 2I + \delta I - 2I - \delta I + \frac{1}{4}I\left(\frac{\delta I}{I}\right)^2$$

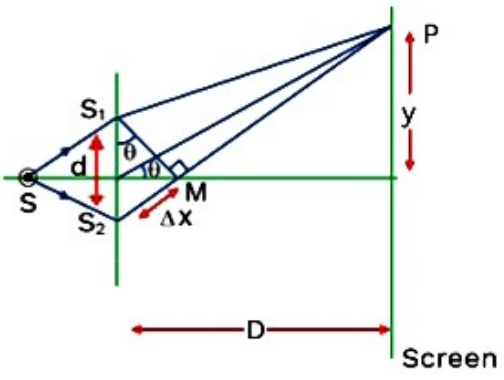
$$\boxed{I_{min} = \frac{(\delta I)^2}{4I}}$$

Young's Double Slit Experiment

Two light sources can never be coherent. So, to produce coherent sources of light, we have to split the light coming from a single source.







Waves from source S_1 and S_2 are interfering at point P on screen.
Here D >> d
So, $S_1P = MP$
Hence, path difference of interfering waves,
$\Delta x = S_2P - S_1P = d \sin\theta$

Phase difference of interfering waves, $\Delta\phi = \frac{2\pi}{\lambda}\Delta x = \frac{2\pi}{\lambda} d\sin\theta$

Intensity of resultant wave on screen,

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

If $I_1 = I_2 = I_0$

$$I = 2I_0(1+\cos\phi) = 4I_0\cos^2\frac{\phi}{2}$$

Maxima will occur when $\cos\phi = 1$

$\phi = 2n\pi \qquad n \in I$

$\theta = 0, 2\pi, 4\pi, 6\pi \ldots\ldots\ldots\ldots$ or

$\Delta x = n\lambda \qquad [\Delta x = 0, \lambda, 2\lambda \ldots\ldots]$

Minima will occur when $\cos\phi = -1$

$\phi = (2n+1)\pi$

$\phi = \pi, 3\pi, 5\pi, 7\pi \ldots\ldots\ldots\ldots$ or

$$\Delta x = (2n+1)\frac{\lambda}{2}$$

$$\Delta x = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \frac{7\lambda}{2} \ldots\ldots\ldots$$

Position of maxima and minima on screen in YSDE

if y << D or θ is very small, then $\sin\theta = \tan\theta = \theta$

$$\Delta x = d\sin\theta = d\tan\theta = d\frac{y}{D}$$

Point P will be a bright spot or maxima if path difference Δx is integral multiple of λ.

$$\Delta x = y\frac{d}{D} = n\lambda$$

Position of nth maxima

$$y_n = n\frac{\lambda D}{d} \qquad \text{where } n = 0,1,2,3 \ldots\ldots\ldots\ldots$$

Bright spots will be obtained at,

$$y = 0, \frac{\lambda D}{d}, \frac{2\lambda D}{d}, \frac{3\lambda D}{d} \ldots\ldots\ldots \text{ from O}$$

Point P will be dark spot or minima if Δx is odd multiple of $\frac{\lambda}{2}$.

$$\Delta x = \frac{yd}{D} = (2n+1)\frac{\lambda}{2} \Rightarrow \text{Position of } n^{th} \text{ minima, } y_n = \frac{(2n+1)\lambda D}{2d} [n = 0, 1, 2, 3 \ldots]$$

Hence, positions of minima are,

$$y = \frac{\lambda D}{2d}, \frac{3\lambda D}{2d}, \frac{5\lambda D}{2d} \ldots\ldots\ldots\ldots \text{ from O.}$$

Q. **In YDSE experiment light of wavelength 589.3 nm is used. The interference pattern is observed on screen which is 100 cm away from the slits. If 10th bright fringe has its centre at a distance of 12 millimetre from the central maximum, calculate the separation between the slits.**

Sol. For the nth maximum fringe, the distance above the central line is

$$X = \frac{n\lambda D}{d}$$

According to the data given, x = 12 mm,
n = 10, λ = 589.3 nm, D = 100 cm.
Thus, the separation between the slits is

$$d = \frac{n\lambda D}{x}$$

$$= \frac{10 \times 589.3 \times 10^{-9} m \times 100 \times 10^{-2} m}{12 \times 10^{-3}}$$

$$= 4.9 \times 10^{-4} \text{ m} = 0.49 \text{ mm.}$$

Q. **A beam of light containing 2 wavelengths, 6500 Å and 5200 Å is used to obtain interference fringes in a YDSE experiment (1 Å = 10^{-10} m). The separation between the slits is 2.0 mm and the screen is placed 120 cm away from the plane of the slits.**

(a) **What is the distance of the 3rd bright fringe formed on the screen from the central maximum for the wavelength 6500 Å.**

(b) **Find the least distance from central maximum where the bright fringes due to both the wavelengths coincide.**

Sol. (a) The centre of the nth bright fringe is at a distance, $y = \frac{n\lambda D}{d}$ from the central maximum.

For the 3rd bright fringe of 6500 Å,

$$y = \frac{3 \times 6500 \times 10^{-10} m \times 1.2 m}{2 \times 10^{-3} m}$$

$= 0.117$ cm ≈ 0.12 cm.

(b) Suppose the mth bright fringe of 6500 Å coincides with the nth bright fringe of 5200 Å.

Then, $\frac{m \times 6500 Å \times D}{d} = \frac{n \times 5200 Å \times D}{d}$

or, $\frac{m}{n} = \frac{5200}{6500} = \frac{4}{5}$

The minimum values of m and n that satisfy this equation are 4 and 5 respectively. The distance of the 4th bright fringe of 6500 Å or the 5th bright fringe of 5200 Å from the central maximum is,

$$y = \frac{4 \times 6500 \times 10^{-10} m \times 1.2 m}{2 \times 10^{-3} m} = 0.156 \text{ cm}$$

≈ 0.16 cm.

Q. **In a YDSE setup, if the screen is moved towards the slits by 5 cm, then the fringe width changes by 3×10^{-5} m. Find the wavelength of light used if separation between slits is 10^{-3} m.**

Sol. The fringe–width β is given by.

$$\beta = \frac{\lambda D}{2d} \quad \text{and} \quad \beta' = \frac{\lambda D'}{2d}$$

where λ is the wavelength of light used, D is the distance of the screen from the two slits and 2d is the separation between two slits.

Now, $\beta - \beta' = \frac{\lambda(D - D')}{2d}$

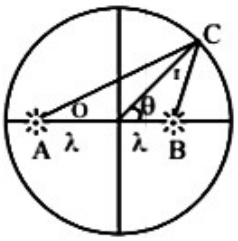
$$\lambda = \frac{(\beta - \beta')2d}{D - D'}$$

$$\lambda = \frac{(3 \times 10^{-5})(10^{-3})}{5 \times 10^{-2}}$$

$= 0.6 \times 10^{-6}$ m $= 6000$ Å

Q. **Two coherent light sources A and B with separation 2λ are placed on the x-axis symmetrically about the origin. They emit light of wavelength λ. Obtain the positions of maxima on a circle of large radius lying in xy-plane and with centre at the origin.**

Sol. From figure,



$$BC = \sqrt{(r^2 + \lambda^2 - 2r\lambda\cos\theta)}$$

and $$AC = \sqrt{(r^2 + \lambda^2 + 2r\lambda\cos\theta)}$$

The path difference between the rays AC and BC reaching point C

$$= AC - BC = n\lambda$$

[$\therefore$ For maximum intensity, path difference should be a whole multiple of wavelength]

Now $$AC = r\left[1 + \frac{\lambda^2}{r^2} + \frac{2\lambda}{r}\cos\theta\right]^{1/2}$$

$$\approx r\left[1 + \frac{2\lambda}{r}\cos\theta\right]^{1/2}$$

$$= r\left[1 + \frac{\lambda}{r}\cos\theta\right]$$

$$= r + \lambda\cos\theta$$

Similarly, $BC \approx r - \lambda\cos\theta$

So, path difference

$$= AC - BC = 2\lambda\cos\theta$$

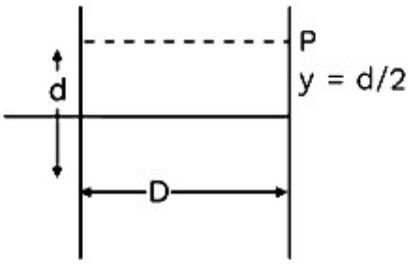
Hence $2\lambda\cos\theta = n\lambda$ or $\cos\theta = n/2$

where $n = 0, 1, 2, 3, ...$

$\therefore$ $\theta = 0°, 60°, 90°, 120°, 180°, 240°, 270°, 300°$

Q. **In a YDSE setup, there is a maxima exactly in front of one slit. What can be the value of possible wavelength?**

Sol.



For maxima, $\Delta x = n\lambda = d\sin\theta = d\tan\theta = \frac{dy}{D} = \frac{d^2}{2D}$

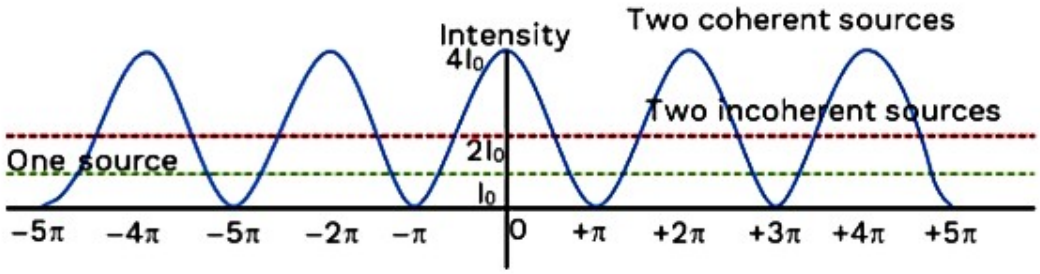
$$\lambda = \frac{d^2}{2nD}, \quad [n = 1, 2, 3...]$$

INTENSITY VARIATION ON SCREEN

If A and I_0 represent amplitude of each wave and the associated intensity on screen, then the resultant intensity at a point on the screen corresponding to the angular position θ as in above figure, is given by,

$$I = I_0 + I_0 + 2\sqrt{I_0^2}\cos\phi = 4I_0\cos^2\left(\frac{\phi}{2}\right), \text{where}, \phi = \frac{2\pi(d\sin\theta)}{\lambda}$$

The intensity variation along the screen near the centre O can be drawn as-



Fringe width (β) : Distance between two consecutive maxima (bright fringes) or minima (dark fringes) is known as fringe width.

For maxima

$$\beta = y_n - y_{n-1} = n\frac{\lambda D}{d} - (n-1)\frac{\lambda D}{d} = \frac{\lambda D}{d}$$

For minima

$$\beta = y_n - y_{n-1} = (2n+1)\frac{\lambda D}{d} - (2(n-1)+1)\frac{\lambda D}{d} = \frac{\lambda D}{d}$$

Note: From here we can say that in YSDE bright and dark fringes are equally spaced.

VARIATION IN FRINGE WIDTH IF WHOLE APPARATUS IS IMMERSED IN A TRANSPARENT MEDIUM:

In YDSE, if whole apparatus is immersed in a liquid of refractive index μ, then fringe width becomes $\frac{1}{\mu}$ times.

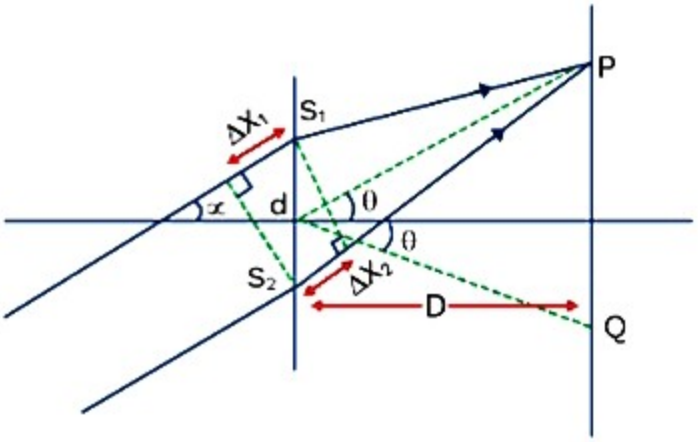
In air, $\lambda = \lambda_0$, $\beta_0 = \frac{\lambda_0 D}{d}$

In medium, $\lambda = \frac{\lambda_0}{\mu}$, $\beta = \frac{\lambda_0}{\mu}\frac{D}{d} = \frac{\beta_0}{\mu}$

YDSE with oblique incidence

When rays interfere at P,

Path difference, $\Delta x = |\Delta x_1 - \Delta x_2| = |d\sin\alpha - d\sin\theta|$

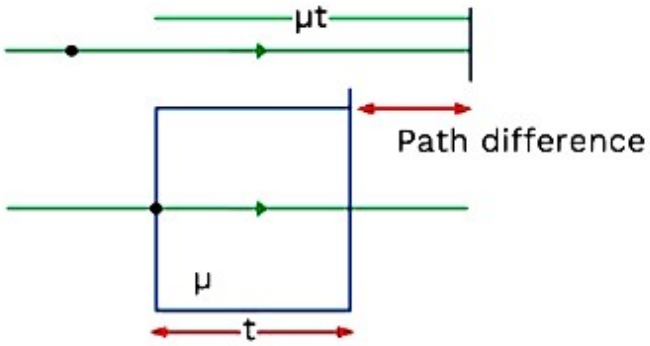


When rays interfere at Q,

Path difference, $\Delta x = |\Delta x_1 + \Delta x_2| = d\sin\alpha + d\sin\theta$

Geometrical path and optical path:

Actual distance travelled by light in medium is called geometrical path. Distance which light has travelled in vacuum in same time for which it is moving in medium is called optical path. For path difference of wave, we should take optical path difference. But if both waves are travelling is same medium then we can take geometrical path difference.



Speed of light in medium, $v = \frac{C}{\mu}$

Time taken to cross slab $= \frac{t}{v} = \frac{\mu t}{C}$

Optical path = Speed of light in vacuum × time taken $= c \times \frac{\mu t}{c} = \mu t$

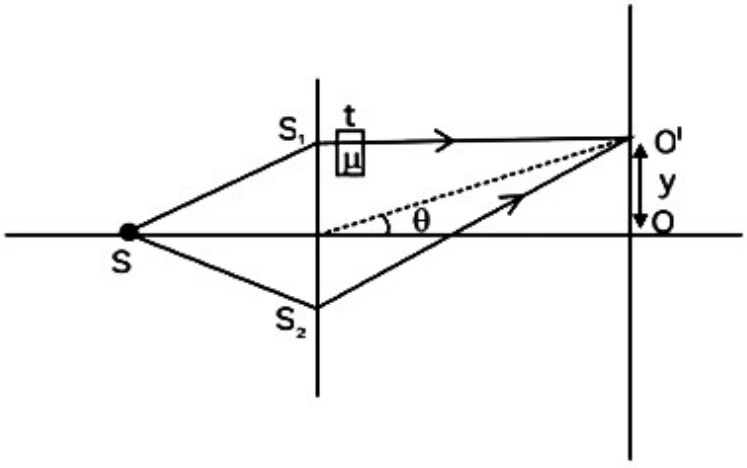
So, here for slab, geometrical path = t

Optical path = μt

Path difference due to slab = Optical path - Geometrical path = μt – t = (μ–1)t.

WHEN A THIN GLASS SLAB IS PUT IN FRONT OF ONE SLIT IN YDSE

If a transparent film of refractive index μ and thickness t is introduced in front of any slit, then central maxima shifts to the side of the slit in which the slab is introduced.



For central maxima, Δx = 0

Path difference due to slab = (μ–1) t

So, at central maxima, $(\mu - 1)t = d\sin\theta = d\tan\theta = \frac{dy}{D}$

Position of central maxima, $y = (\mu - 1)\frac{tD}{d}$

Here all fringes will shift by same distance upwards.

Q. **Monochromatic light with λ =600 nm is used in a YDSE experiment. How many fringes will shift when a transparent sheet of refractive index 1.6 with thickness 1.8×10^{-5} m is made to cover one of the slits?**

Sol. As derived earlier, the total fringe shift = $\frac{w}{\lambda}(\mu - 1)t$.

As each fringe width = w,

The number of fringes that will shift = $\frac{\text{total fringe shift}}{\text{fringe width}}$

$$= \frac{\frac{w}{\lambda}(\mu - 1)t}{w} = \frac{(\mu - 1)t}{\lambda} = \frac{(1.6 - 1) \times 1.8 \times 10^{-5}\ m}{600 \times 10^{-9}\ m} = 18$$

Q. **The intensity at the position of formation of central maximum remains changed when a glass plate of thickness t is placed in path of one of the interfering beams in a YDSE experiment. If λ is wavelength of light used, then what is the minimum thickness of the glass plate?**

Sol. For minimum thickness $(n-1)t = \lambda$
$t = \lambda / n-1 = \lambda / 1.5-1 = 2\lambda$

Q. **In double slit experiment, a glass plate of refractive index 1.5 of thickness t is introduced in the path of one of the interfering beams of wavelength λ. The position of central maxima has got shifted by $\frac{2\lambda D}{d}$. Then, what would be the thickness of plate?**

(A) 4λ **(B) 2λ**
(C) λ **(D) 3λ**

Sol. $\Delta y = D(\mu-1)t/d$ ...(1)

$\Delta y = 2\lambda D/d$ ...(2)

From equation 1 & 2

$D(\mu-1)t/d = 2\lambda D/d$

$t = 2\lambda/(\mu-1)$

$t = 2\lambda/1.5-1$

$t = 4\lambda$

Q. **In a Young's double slit experiment, the separation between slits is 2×10^{-3} m whereas the distance of screen from the slits is 2.5 m. Light of wavelengths in the range of 2000-8000 Å is allowed to fall on slits. Find the wavelengths in the visible region that will be present on the screen at 10^{-3} m from the central maxima. Also, find the wavelengths that will be present at the point of screen in the infrared as well as in the ultraviolet region.**

Sol. The distance of the bright fringes from the central maximum on the screen is given by

$$x = n\frac{\lambda D}{2d},$$

$n = 0, 1, 2, 3.....$

where n = 0, corresponds to the central maximum.

$$\therefore \quad \lambda = \frac{x \times (2d)}{nD}$$

$$\text{For } n = 1,\ l1 = \frac{(10^{-3})(2 \times 10^{-3})}{1 \times (2.5)}$$

= 8000 Å (infrared)

$$\text{For } n = 2,\ l2 = \frac{(10^{-3})(2 \times 10^{-3})}{2 \times 2.5}$$

= 4000 Å (visible)

$$\text{For } n = 3,\ l3 = \frac{(10^{-3})(2 \times 10^{-3})}{3 \times 2.5}$$

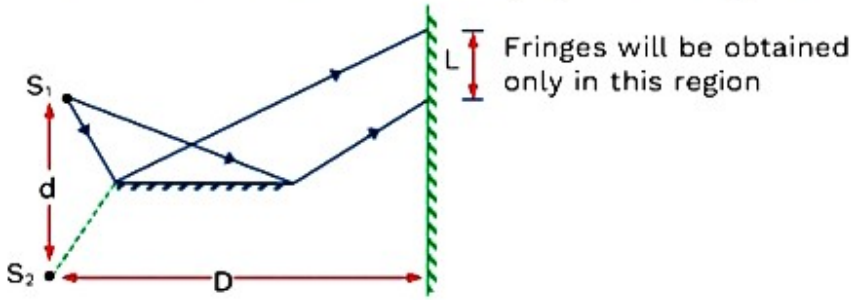
= 2666 Å (ultraviolet)

$$\text{For } n = 4,\ \lambda 4 = \frac{(10^{-3})(2 \times 10^{-3})}{4 \times 2.5}$$

= 2000 Å (ultraviolet)

SIMILAR PATTERNS LIKE YDSE

Lloyd's mirror: In this setup, coherent source is mirror image of source. When light is reflected from a denser medium then there is abrupt phase change of π.

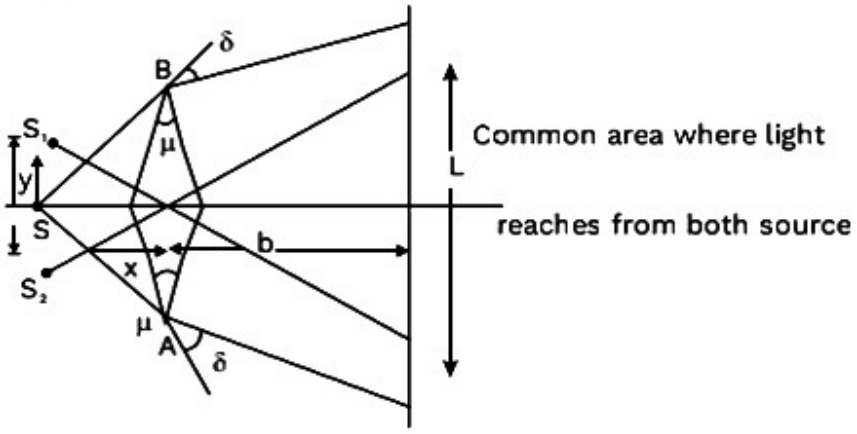


Here, interference will occur due to direct light from source and reflected light from mirror.

Fringe width $\beta = \frac{\lambda D}{d}$

Fresnel Biprism:

In this setup, two thin prisms are used to create coherent sources



$d = 2y = x\delta = x(\mu - 1)A$

$D = x + b$

$\beta = \frac{\lambda D}{d} = \frac{\lambda(x + b)}{2x(\mu - 1)A}$

Q. **In Fresnel's biprism (μ = 1.5) experiment, the distance between source and biprism is 0.3 m and that between biprism and screen is 0.7 m and angle of prism is 1°. The fringe width when λ = 6000 Å will be.**

(A) 4 cm **(B) 5cm**

(C) 0.1 cm **(D) 0.011 cm**

Sol. $\beta = \dfrac{(a+b)\lambda}{2a(\mu-1)\alpha}$

where a = distance between source and biprism = 0.3 m

b = distance between biprism and screen = 0.7 m.

a = Angle of prism = 1°, μ = 1.5,

$\lambda = 6000 \times 10^{-10}$m

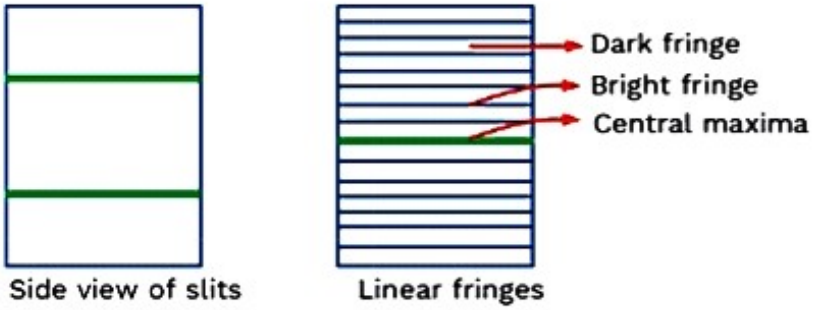
Hence,

$$\beta = \frac{(0.3+0.7)\times 6\times 10^{-7}}{2\times 0.3(1.5-1)\times\left(1^{\circ}\times\dfrac{\pi}{180}\right)}$$

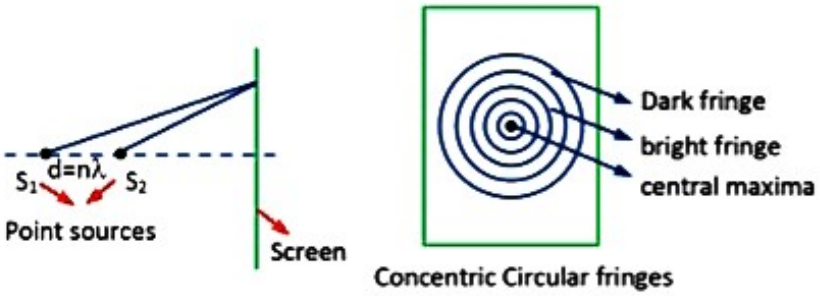
$= 1.14 \times 10^{-4}$m = 0.0114 cm

SHAPE OF INTERFERENCE PATTERN:

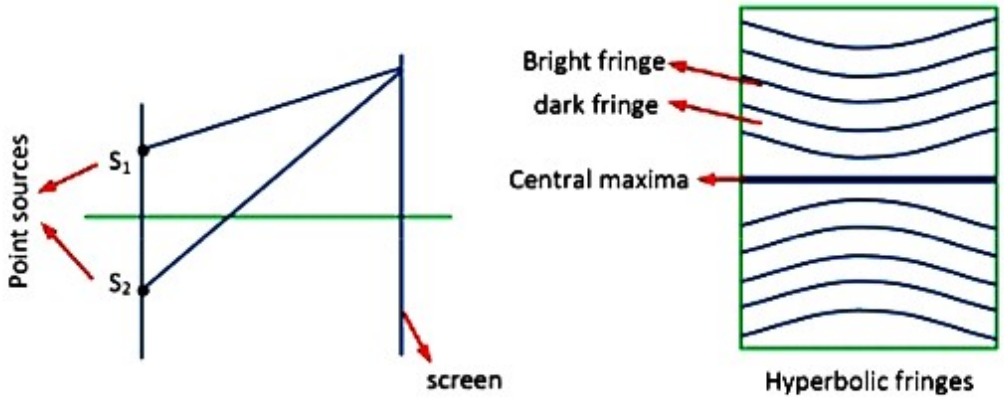
1. When interference is due to 2 linear slits



2. When interference is due to 2 point sources kept on a line perpendicular to screen



3. When interference is due to 2 point sources kept on a line parallel to screen



YDSE WITH WHITE LIGHT

The position of central maxima does not depend upon the wavelength of the light used. We get a white spot at the position of central maxima when white light is used to find the central maxim in YDSE because only at the position of central maxima, all the wavelengths contained in white light will show constructive interference.
Slightly above or slightly below the central maxima, we will get colored fringes.

VIBGYOR



On moving away from the central maxima, first maxima & minima have violet color but in the nearby region of the central maxima, reddish color is more dominant because in this region, the intensity of the violet color decreases at a faster rate as compared to the red color.

In an usual YDSE setup with monochromatic light source, we get a large number of identical interference fringes. So, it becomes difficult to locate the position of central maxima.

So, the experiment is done with white light to determine the position of central maxima in such cases.

Note: When a light wave is reflected from a denser medium, there is an abrupt phase change of π and no phase change occurs when reflection takes place from rarer medium.

Q. **White light is being used in a double slit experiment and interference pattern is being observed on a screen kept 2.5 m away. The slit separation is given to be 0.5 mm. At 2.0 mm from the central white fringe, the first violet fringe is formed and at 3.5 mm from the central white fringe, the first red fringe is formed. Find the wavelengths of the red and violet light.**

Sol. For the first bright fringe, the distance from the centre is,

$$y = \frac{D\lambda}{d}$$

For violet light,

$$y = 2.0 \text{ mm}.$$

Thus,

$$2.0 \text{ mm} = \frac{(2.5\text{m})\lambda}{0.5\text{mm}}$$

or,

$$\lambda = \frac{(0.5\text{mm})(2.0\text{mm})}{2.5}$$

$$= 400 \text{ nm}.$$

Similarly, for red light, y = 3.5 mm.

Thus,

$$3.5 \text{ mm} = \frac{(0.2\text{m})\lambda}{0.5\text{mm}}$$

or,

$$\lambda = 700 \text{ nm}.$$

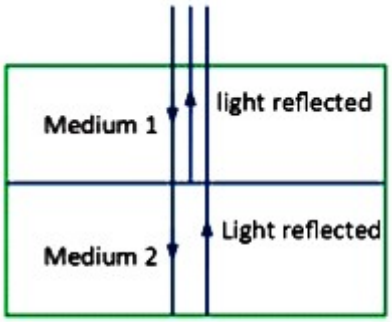
THIN FILM INTERFERENCE:

When light passes the boundary between two transparent media, some light is reflected at the boundary & some is refracted through the boundary. As shown in the figure, some light is reflected from first surface & some from second surface.

If we consider a monochromatic incident light, the two reflected waves are also monochromatic incident light wave via amplitude division. These waves interfere since they are superposed along the same normal line.

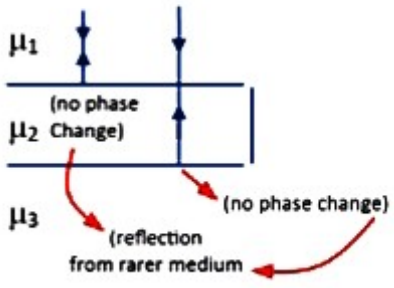
The phase difference between two waves arises due to

1. Optical path difference (due to distance travelled)
2. Reflection from a denser medium (the second factor is irrelevant for reflection at rarer medium.)



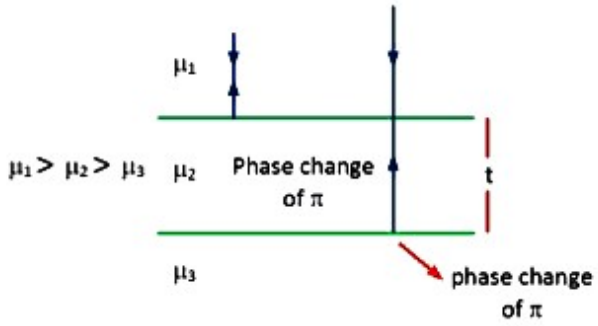
Three situations may arise-

1. Neither wave experience a phase change upon reflection ($\mu_1 > \mu_2 > \mu_3$).



$$\Delta x = \frac{2\mu_2 t}{\mu_1}$$

2. Both the waves suffer a phase change upon reflection.



$$\Delta x = \frac{2\mu_2}{\mu_1} t$$

In either of the above cases, the phase change due to reflection is irrelevant as there is no phase change due to reflection. In either of these cases, phase change is determined only from path difference.

Condition for constructive Interference,

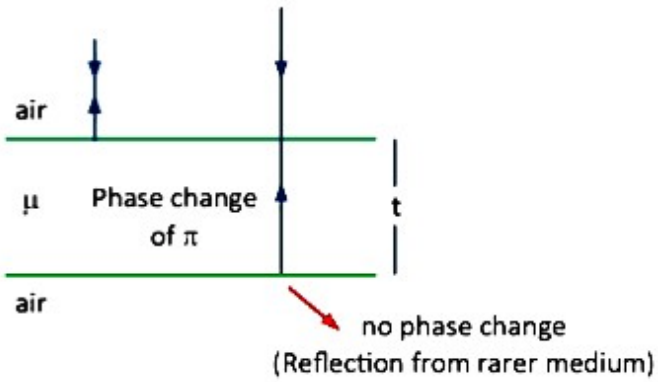
$$\frac{2\mu_2}{\mu_1} t = n\lambda$$

Condition for destructive interference,

$$\frac{2\mu_2}{\mu_1} t = \left(n + \frac{1}{2}\right)\lambda$$

3. One of the reflected waves experience a phase change of π radian upon reflection & the other wave does not.

$$\Delta x = 2\mu t - \frac{\lambda}{2}$$



Due to phase change of π(path change of $\frac{\lambda}{2}$), the conditions are reversed.

$2\mu t = n\lambda$ (for destructive interference)

$2\mu t = \left(n + \frac{1}{2}\right)\lambda$ (for constructive interference)

Q. **For light incident on a soap film having thickness 5×10^{-5} cm, maximum wavelength reflected in the visible region is equal to 5320 Å. Find the refractive index of the film.**

Sol.

$$2\mu t \cos r = (2n+1)\frac{\lambda}{2}$$

or $$\mu = \frac{(2n+1)}{2t\cos r}\left(\frac{\lambda}{2}\right) = \frac{(2n+1)\times 5320\times 10^{-10}}{2\times 5\times 10^{-5}\times 10^{-2}\times 1}$$

$\Rightarrow \mu = 1.33$

Q. **White light is incident normally on a glass plate of thickness 0.50×10^{-6} m and index of refraction 1.50. Which wavelengths in the visible region (400 nm - 700 nm) are strongly reflected by the plate?**

Sol. The light of wavelength λ is strongly reflected if the light rays reflected are interfering constructively.

As we know, the condition for constructive interference

$$2\mu t = \left(n+\frac{1}{2}\right)\lambda.$$

Here, $2\mu t = 2 \times 1.5 \times (0.5 \times 10^{-6})m = 1.5 \times 10^{-6}$ m.

Putting $\lambda = 400$ nm, $1.5 \times 10^{-6} = \left(n+\frac{1}{2}\right)400 \times 10^{-9} \Rightarrow n = 3.25$

Similarly, by putting, $\lambda = 700$ nm.

$$1.5 \times 10^{-6} = \left(n+\frac{1}{2}\right)(700 \times 10^{-9})$$

$\Rightarrow \quad n = 1.66$

Thus, within 400 nm to 700 nm, for integral values of n = 2 and 3.

Now, $\lambda = \frac{4\mu t}{2n+1} = 600\,\text{nm}$ and 429 nm are strongly reflected.

Diffraction

Diffraction

(a) Diffraction is bending or spreading of waves that encounter an object (a barrier or an opening) in their path.

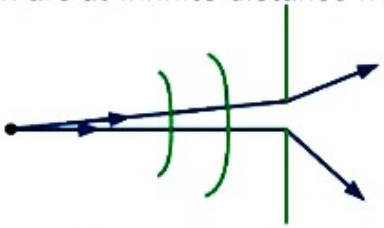
Diffraction is studied in two parts according to distance between source and screen

(i) Fresnel diffraction

The source and/or screen are at a finite distance from the aperture.

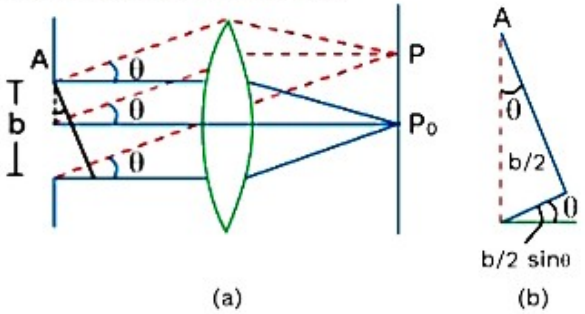
(ii) Fraunhofer diffraction

The source and screen are at infinite distance from the diffracting aperture.

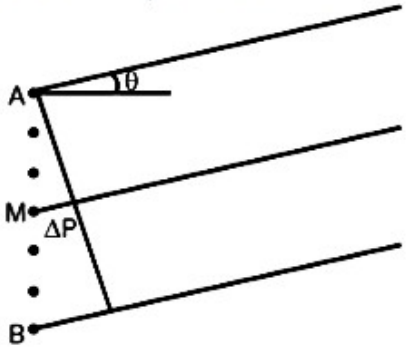


Fraunhofer is a special case of Fresnel diffraction in which source and screen distance are infinite.

FRAUNHOFER DIFFRACTION BY A SINGLE SLIT



When parallel beam of coherent source of light incident normally on a slit having width b, then every point on secondary wavelets act as a source, according to Huygen's principle lights from these different sources produce interference on a screen away from this slit. In order to analyse this into pattern let us consider a point P on screen at an angle θ with respect to the center point on the screen. Now path difference for the point P from the two source one at the extreme point and mid-point would be b/2 sinθ



Here $\Delta P = \frac{b}{2}\sin\theta$.

If for any point P, $\frac{b}{2}\sin\theta = \frac{\lambda}{2}$ the phase difference between extreme point A and mid-point M would be π and they interfere destructively. Moreover for any point in upper section, there exist a point in lower section which are in opposite phase cancelling each other and forming minima for this point P.

But if for any point P, b $\sin\theta = \frac{3\lambda}{2}$ then we have to divide the slit in 3 section which are in opposite phase, In which light from the 2-sections cancel each other but light from the other 1-section remains uncancelled producing maxima for this condition

So, condition for minima would be $\frac{b}{2}\sin\theta = \frac{\lambda}{2}$

or, $b \sin\theta = \lambda$ (first minimum).

Similar arguments show that other minima (zero intensity) are located at points corresponding to

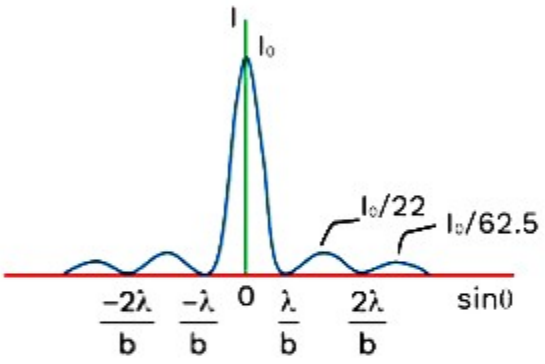
$b\sin\theta = 2\lambda, 3\lambda.....$

or, $b \sin\theta = n\lambda$ (dark fringe).

$$I = I_0 \frac{\sin^2\beta}{\beta^2}$$

Here $\beta = \pi b \sin\theta/\lambda$

VARIATION OF INTENSITY AS A FUNCTION OF sinθ



Most of the diffracted light is distributed between $\sin\theta = -\frac{\lambda}{b}$ and $+\frac{\lambda}{b}$. The intensity at the first maximum after the central one is only 1/22 of the intensity of the central maximum.

The divergence in θ is inversely proportional to the width b of the slit. If the slit-width b is decreased, the divergence is increased and the light is diffracted in a wider cone.
On the other hand, if the slit-width is much larger compared to the wavelength, $\lambda/b = 0$ and the light continue undiffracted in the direction $\theta = 0$.
This clearly indicates that diffraction effects are observable only when the obstacle or the opening has dimensions comparable to the wavelength of the wave.

Q. **What is the angular width of the central maxima when a light of wavelength 6328 Å is made incident on a slit of width 0.2 mm normally.**

Sol. The angular width of central maxima is,

$$2\theta = 2\frac{\lambda}{a} = \frac{2 \times 6328 \times 10^{-10}}{2 \times 10^{-4}} \text{ radian.}$$

$$= 6328 \times 10^{-6} \times \frac{180}{\pi}$$

Degree = 0.36°

Q. **A plane wave front ($\lambda = 6 \times 10^{-7}$m) is falling on a slit which is 0.4 mm wide. A convex lens having a focal length of 0.8 m is placed behind the slit. This lens focusses the light on a screen. What is the linear diameter of second maximum?**

(A) 6 mm **(B) 12 mm** **(C) 3 mm** **(D) 9 mm**

Sol. For secondary maxima $d \sin\theta = \frac{5\lambda}{2}$

$$\Rightarrow d\theta = d \cdot \frac{x}{D(\approx f)} = \frac{5\lambda}{2}$$

$$\Rightarrow 2x = \frac{5\lambda f}{d} = \frac{5 \times 0.8 \times 10^{-7}}{4 \times 10^{-4}} = 6 \times 10^{-3}\,m = 6mm$$

Q. **A light beam which is parallel and monochromatic beam is made incident normally on a narrow slit. A diffraction pattern is observed on a screen kept perpendicularly to the incident beam direction. What is the phase difference between the rays coming from the edges of the slit at the first maximum of the diffraction pattern.**

(A) 0 **(B)** $\frac{\pi}{2}$

(C) π **(D)** 2π

Sol. The phase difference between the rays coming from the two edges of the slit is given by the formula.

$$\frac{2\pi}{\lambda} d \sin\theta$$

where, d is the width of the slit.

the first minimum occurs when the following condition is satisfied.

$$\sin\theta = \frac{\lambda}{d}$$

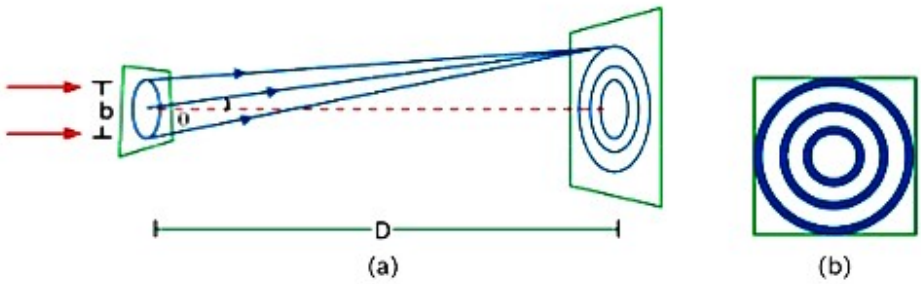
$$\therefore \frac{2\pi}{\lambda} d \frac{\lambda}{d} = 2\pi$$

FRAUNHOFER DIFFRACTION BY A CIRCULAR APERTURE

On passing a light beam through a circular hole in a solid plate, there is diffraction of the light beam through the hole. The diffraction pattern appears in the shape of alternate bright and dark fringes on a screen placed in front of the hole at some distance from it.
The alternate bright and dark fringes are due to the interference between secondary wavelets emitted from the hole on the plate.
The mathematical analysis shows that the first dark ring is formed by the light diffracted from the hole at an angle θ with the axis where,

$$\sin\theta = 1.22 \frac{\lambda}{b}$$

For rectangular slit, for first minima, $\sin\theta = \frac{\lambda}{b}$



Here λ is the wavelength of the light used and b is the diameter of the hole. If the screen is at a distance D (D >> b) from the hole, the radius of the first dark ring is,

$$R \approx 1{\cdot}22\frac{\lambda D}{b}$$

On converging the transmitted light from the hole by using a lens, the first dark ring formed has a radius given by,

$$R = 1{\cdot}22\frac{\lambda f}{b}$$

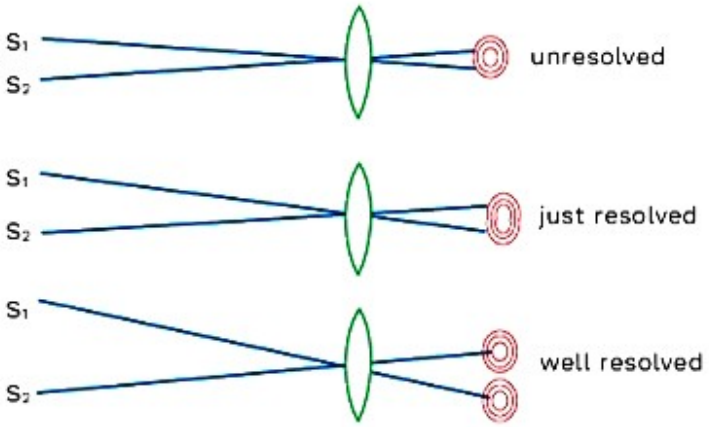
A large part of the light from the hole is focussed in the first dark ring. Radius of the first dark ring is known as the radius of diffraction disc.

From the above figure, it can also be found that,

$$\sin\theta \approx \frac{1{\cdot}22\lambda}{b}$$

LIMIT OF RESOLUTION

The fact that a lens forms a disc image of a point source, puts a limit on resolving two neighbouring points imaged by a lens.



For two point ight sources S_1 and S_2 kept in front of a converging lens, if the separation between the centres of the image-discs is small in comparison to the radii of the discs, the discs will largely overlap on one another and it will appear like a single disc.

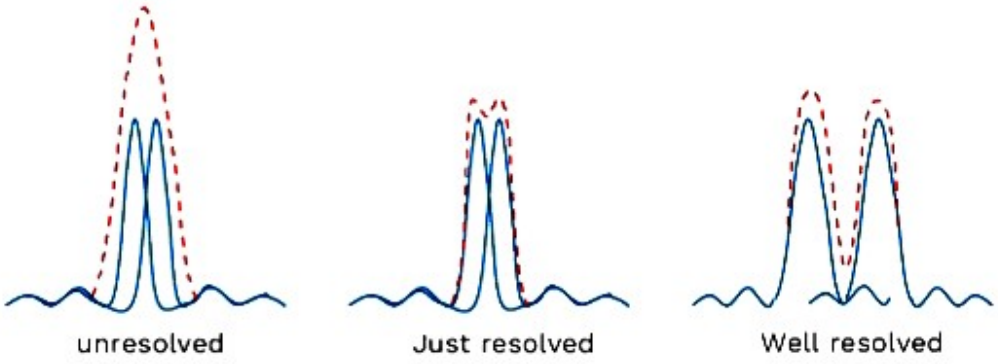
As long as they appear as a single disc, we say that the points are unresolved.

If the separation between them is increased to a suitable value, it becomes easy to distinguish between the two discs in the image formed. At that separation, the sources are just resolved.

The angular radius θ of the diffraction disc is given by $\sin\theta = \frac{1.22\lambda}{b}$, where b is the diameter of the lens.

RAYLEIGH CRITERION

Whether two-disc images of nearby points are resolved or not may depend on the person viewing the images. Rayleigh suggested a quantitative criterion for resolution.
When the centre of a bright disc lies on the periphery of another bright disc in the images, then the images are said to be "just resolved".

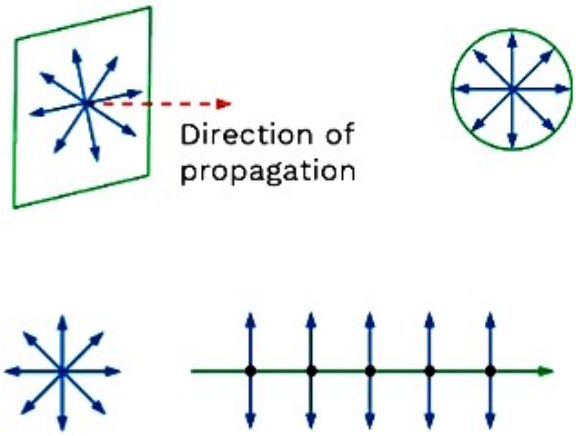


Polarisation

Light propagates in the form of transverse electromagnetic waves. The magnitude of electric field is much larger as compared to magnitude of magnetic field. It is normally preferred to describe light as electric field oscillations.

Unpolarised light

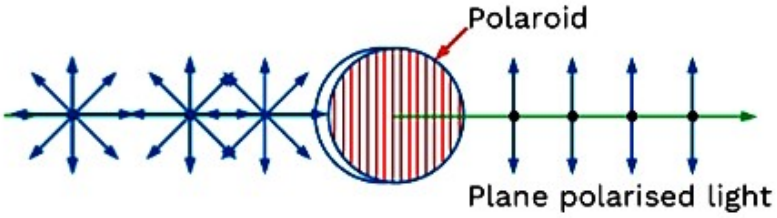
When the electric field oscillations of a light are in all the directions in the plane perpendicular to the direction of propagation of light, then the light is called as unpolarised light.



If the electric field oscillations in a light beam are in one direction only, then it is said to be polarised light.
The plane which is perpendicular to the plane of oscillation is called plane of polarisation. Light can be polarised by transmitting through certain crystals such as tourmaline or polaroids.

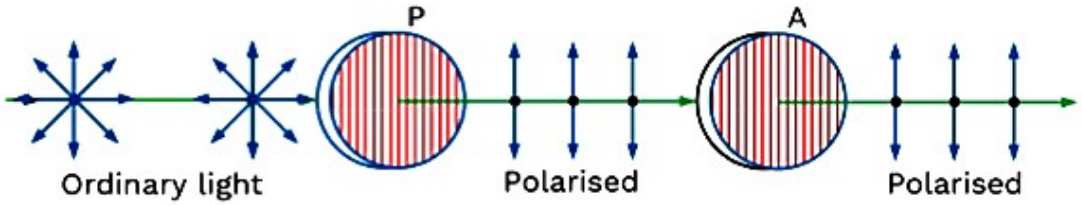
Polaroids

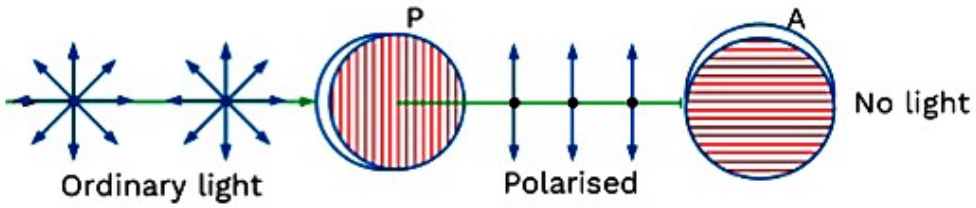
Polaroids are used to produce plane polarised light. They work on selective absorption principle and are more effective as compared to tourmaline crystals.
It is a thin film of ultramicroscopic crystals of iodoquinine sulphate with their optic axis parallel to each other.



(i) Polaroids allow the light oscillations parallel to the transmission axis pass through them.

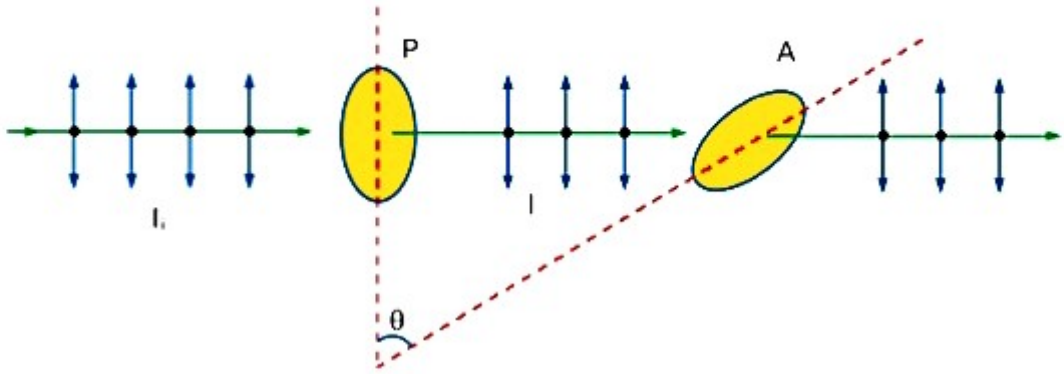
(ii) The crystal on which unpolarised light is incident is called polariser. Crystal or polaroid on which polarised light is incident is called analyser.





MALUS LAW

As per Malus law, the intensity of the polarised light transmitted through the analyser varies as the square of the cosine of the angle between the plane of transmission of the analyser and the plane of the analyser and the plane of polariser.



(i) $I = I_0 \cos^2\theta$ and $A^2 = A_0^2 \cos^2\theta \Rightarrow A = A_0 \cos\theta$

If $\theta = 0°$, $I = I_0$, $A = A_0$, If $\theta = 45°$, $I = \frac{I_0}{2}$, $A = \frac{A_0}{\sqrt{2}}$, If $\theta = 90°$, $I = 0$, $A = 0$

(ii) If I_i = Intensity of unpolarised light.

So, $I = \frac{I_i}{2}$, i.e., if an unpolarised light is converted into plane polarised light (say by passing it through a polaroid or a Nicol-prism), its intensity becomes half.

Note: Percentage of polarisation $= \frac{(I_{max} - I_{min})}{(I_{max} + I_{min})} \times 100$

Q. **Light has amplitude A and angle between analyser and polariser is 60°. Light reflected by analyser has amplitude,**

(A) $A\sqrt{2}$ **(B)** $A / \sqrt{2}$

(c) $\sqrt{3}A / 2$ **(d) A/2**

Sol. We know, amplitude of light reflected by analyser, A' = Acosθ
where, A is amplitude of light and θ is angle between analyser and polariser.
so, A' = Acos60° = A/2
hence, option (D) is correct.

Q. **A polaroid is placed at 45° to an incoming light of intensity I_0. Now the intensity of light passing through polaroid after polarisation would be.**

(A) I_0 **(B)** $I_0/2$

(C) $I_0/4$ **(D) Zero**

Sol. $I = I_0 \cos^2\theta$

$I = I_0/2$

Q. **Unpolarized light of intensity 32Wm^{-2} passes through three polarizers such that transmission axes of the first and second polarizer makes an angle 30° with each other and the transmission axis of the last polarizer is crossed with that of the first. The intensity of final emerging light will be.**

(A) 32 Wm^{-2} **(B) 3 Wm^{-2}**

(C) 8 Wm^{-2} **(D) 4 Wm^{-2}**

Sol. Angle between P_1 and P_2 = 30°(given)

Angle between P_2 and P_3 = θ = 90° – 30° = 60°

The intensity of light transmitted by P_1 is.

$$I_1 = \frac{I_0}{2} = \frac{32}{2} = 16\,\frac{W}{m^2}$$

According to Malus law, the intensity of light transmitted by P_2 is,

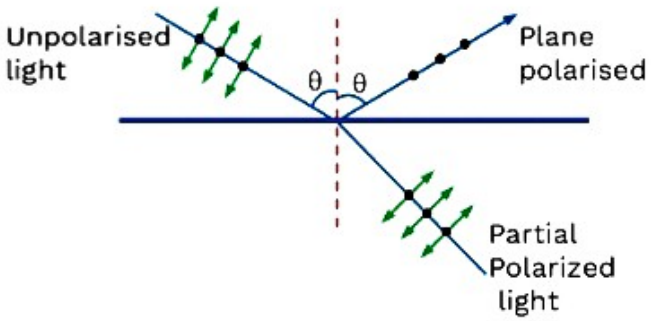
$$I_2 = I_1 \cos^2 30^\circ = 16\left(\frac{\sqrt{3}}{2}\right)^2 = 12\,\frac{W}{m^2}$$

Similarly, intensity of light transmitted by P_3 is,

$$I_3 = I_2 \cos^2\theta = 12\cos^2 60^\circ = 12\left(\frac{1}{2}\right)^2 = 3\,\frac{W}{m^2}$$

BREWSTER'S LAW

'At the condition of maximum polarization, the incident rays must fall at such an angle that reflected rays and refracted rays are perpendicular to each other'.



Also, $\mu = \tan\theta_p$. This relation is Brewster's law.

(i) For $i < \theta_p$ or $i > \theta_p$
Both reflected and refracted rays become partially polarized.

(ii) For glass $\theta_p \approx 57°$, for water $\theta_p \approx 53°$

Q. **The angle of polarisation for any medium is 60°, what will be critical angle for this.**

(A) $\sin^{-1}\sqrt{3}$ **(B)** $\tan^{-1}\sqrt{3}$

(C) $\cos^{-1}\sqrt{3}$ **(D)** $\sin^{-1}\frac{1}{\sqrt{3}}$

Sol. $\mu = \tan(\theta) = \tan 60 = \sqrt{3}$

By Snell's law, $\sqrt{3}\sin(\theta) = 1\sin 90$

$\sin\theta = 1/\sqrt{3}$

$\theta = \sin^{-1}(1/\sqrt{3})$

Q. **The angle of incidence at which reflected light is totally polarized for reflection from air to glass (refraction index n) is.**

(A) $\sin^{-1}(n)$ **(B)** $\sin^{-1}\left(\frac{1}{n}\right)$

(C) $\tan^{-1}\left(\frac{1}{n}\right)$ **(D)** $\tan^{-1}(n)$

Sol. (D)

According to Brewster's law of polarization, $n = \tan(i_p)$

where i_p is angle of incidence.

Therefore, $i_p = \tan^{-1}(n)$

Q. **A ray of unpolarised light is incident on a glass plate of refractive index 1.54 at polarising angle, then angle of refraction is.**

(A) 33° **(B) 44°**

(C) 57° **(D) 90°**

Sol. (A)

$\mu = \tan i_p$ or $\tan i_p = 1.54$

i.e., $i_p = \tan^{-1} 1.54 = 57°$,

But, $r + i_p = 90°$

$\therefore \quad r = 90° - i_p$

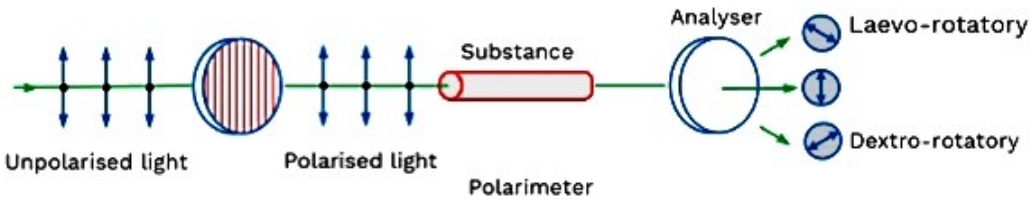
$= 90° - 57° = 33°$.

OPTICAL ACTIVITY AND SPECIFIC ROTATION

There are certain substances which rotate the plane of polarisation of light about the direction of propagation of light by a certain angle when plane polarised light is passed through them. It is called optical rotation or optical activity and these type of substance are called optically active substances.

If the plane of polarisation is rotated clockwise (on looking opposite to the direction of light), then the optical active substance is a dextro-rotatory or right handed substance.

If the substance rotates the plane of polarization anti-clockwise, it is called laevo-rotatory or left handed.



A polarimeter is used to measure the optical activity of a substance in terms of 'specific rotation' which is defined as the rotation produced by a solution of length 10 cm (1 dm) and of unit concentration (i.e., 1g/cc) for a given wavelength of light at a given temperature,

i.e, $[\alpha]^{\lambda}_{t°C} = \dfrac{\theta}{L \times C}$ where θ is the rotation in length L at concentration C.

Q. **Suppose there is one litre water and 80 gram of impure sugar dissolved in it. This solution gives an optical rotation of 9.9. When it is placed in 20 cm long tube. If can concentration of sugar solution is 75 gram litre, then specific rotation of sugar would be.**

(A) 44° **(B) 55°**

(C) 66° **(D) 73°**

Sol. (C)

$$c = \frac{\theta}{lS} \text{ or } S = \frac{\theta}{ic} = \frac{9.9}{2 \times 0.075} \Rightarrow S = 66°.$$

Applications and uses of polarization

(1) The refractive index of any transparent substance can be determined by determining the polarizing angle.

Use Brewster's Law:

$$\boxed{\mu = \tan i_p}$$

(2) The intensity of strong light that hurts your eyes can be reduced by sing polarization

(3) In LCD T.V., Wates, Calculators uses liquid crystal to rotate polarized light.

(4) In polar region we uses compass for navigation. Solar compass uses polarisation of scattered sunlight.

(5) Polarisation is used to find asymmetries in optically active substance..